

A simplified numerical-experimental framework for seismic fragility assessment of high-voltage transformers

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High-voltage electrical substations are critical components of power transmission networks, and their seismic damage may compromise the continuity of service after major earthquakes. Among substation equipment, inductive voltage transformers are particularly vulnerable because of their slender geometry, non-uniform mass distribution and the presence of brittle insulating components.

This work presents a simplified numerical-experimental procedure for the seismic assessment of high-voltage voltage transformers. Two real case studies, representative of the same structural typology but characterized by different geometrical dimensions, mass distributions and voltage levels, are considered. Starting from shake-table experimental tests, a reduced-order finite element model with lumped masses is developed to reproduce the global dynamic response of the devices through a limited number of equivalent mechanical parameters.

The model is calibrated through an inverse optimization procedure based on dimensionless parameters describing stiffness, mass and damping properties. Both acceleration response and base bending moments are used as target quantities, allowing a balanced representation of the dynamic behaviour. The calibrated model is then adopted to derive seismic fragility curves using Incremental Dynamic Analysis, Multiple Stripe Analysis and Endurance Time analysis.

Peak Ground Acceleration is selected as the intensity measure, while collapse is defined by the exceedance of a critical base bending moment, consistently with the brittle behaviour of insulating components [1-2]. The results show that the proposed reduced-order model can reproduce the experimental response of the investigated transformers and can be effectively used for fragility assessment. The comparison among the adopted dynamic procedures highlights the influence of the analysis method on the estimated fragility and confirms the potential of calibrated simplified models for rapid seismic vulnerability assessment of high-voltage electrical equipment.

References

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